# Advanced Thermodynamic Diagnostics in HVAC Systems: Utilizing Pressure-Enthalpy Analysis and Saturation Relationships

The transition from rudimentary diagnostic practices to advanced thermodynamic analysis marks the definitive boundary between component-level parts replacement and true system engineering. For decades, the heating, ventilation, and air conditioning (HVAC) and commercial refrigeration industries relied heavily on heuristic rules of thumb. Technicians evaluated equipment health based primarily on isolated gauge pressures or gross temperature differentials. However, as environmental regulations force the adoption of sophisticated, low-Global Warming Potential (GWP) refrigerants and as energy efficiency mandates drive the ubiquitous integration of variable-speed, inverter-driven compressors, these legacy methods have become severely inadequate. To properly evaluate modern refrigeration circuits, one must visualize the entire thermodynamic process.

The Pressure-Enthalpy (P-H) diagram, historically reserved for design engineers in laboratory environments, provides this precise visual and mathematical map.[1] By plotting the actual operating cycle against the theoretical ideal vapor-compression cycle on a P-H diagram, analysts can mathematically and visually pinpoint inefficiencies, capacity losses, and mechanical degradation that standard gauge readings obscure. This comprehensive report details the application of pressure-temperature relationships and P-H diagrams as foundational maintenance diagnostic tools. It provides an exhaustive examination of saturation fundamentals, modern refrigerant characteristics, the complexities of temperature glide in zeotropic blends, specific thermodynamic fault signatures, the superheat and subcooling diagnostic matrix, and the complex considerations required when evaluating inverter-driven systems.

## The Foundations of Thermodynamics and Phase Change

Before plotting a Pressure-Enthalpy diagram or interpreting system performance anomalies, an absolute mastery of saturation fundamentals is required. The entire mechanical vapor-compression refrigeration cycle is predicated on the absorption and rejection of latent heat during phase changes.[3] The conditions under which these critical phase changes occur are dictated entirely by the pressure-temperature relationship of the specific refrigerant fluid employed in the circuit.

Within a closed refrigeration system, the working fluid continuously transitions between liquid and vapor states. The thermodynamic point at which these two phases coexist in absolute equilibrium is known as saturation. Heat added to a saturated liquid will boil it into a vapor, utilizing the latent heat of vaporization. Conversely, heat removed from a saturated vapor will condense it back into a liquid, utilizing the latent heat of condensation.[3] Because this process occurs at a constant pressure and temperature for pure fluids, knowing the pressure of a saturated refrigerant instantly yields its temperature, and vice versa. The capacity of a fluid to absorb heat during a phase change dwarfs its capacity to absorb heat when simply changing temperature. For example, utilizing the thermodynamic properties of water as an analogy, it requires merely 142 British Thermal Units (BTUs) to heat one pound of water from standard room temperature to its boiling point of 212°F, representing sensible heat. However, to convert that same pound of 212°F water into 212°F steam requires an additional 970 BTUs of latent heat.[3] This massive disparity underscores why the refrigeration cycle relies on phase changes in the evaporator and condenser to move thermal energy.

Once the refrigerant has completely boiled into a 100% vapor state within the evaporator coil, any additional thermal energy absorbed from the conditioned space raises the temperature of the vapor above its saturation point. This sensible heat addition is termed superheat.[3] Ensuring adequate superheat is a critical mechanical safeguard to verify that only pure vapor enters the suction port of the compressor, thereby precluding the catastrophic hydraulic damage known as liquid slugging, which can shatter valves and destroy internal scroll sets.[4]

Conversely, once the refrigerant has fully condensed into a 100% liquid state within the condenser coil, further removal of thermal energy lowers its temperature below the saturation point. This sensible heat reduction is termed subcooling.[3] Proper subcooling guarantees that a solid, continuous column of liquid refrigerant reaches the thermostatic expansion valve (TXV) or electronic expansion valve (EEV). Without adequate subcooling, the liquid refrigerant may flash into a vapor prematurely in the liquid line, a phenomenon that severely limits the mass flow rate through the metering device and drastically reduces the overall cooling capacity of the system.

## Modern Refrigerant Dynamics and Fluid Classifications

The global regulatory shift away from ozone-depleting legacy refrigerants like R-22 has introduced a complex matrix of alternatives, heavily influenced by mandates such as the American Innovation and Manufacturing (AIM) Act.[7] These replacement refrigerants fall into distinct categories based on their chemical compositions, which directly influence their pressure-temperature behaviors and safety handling protocols.

Pure refrigerants and azeotropic mixtures behave uniformly during phase changes. Pure refrigerants, such as R-134a, possess a single, distinct saturation temperature for any given pressure.[8] R-134a, technically known as 1,1,1,2-Tetrafluoroethane, is heavily utilized in medium-temperature commercial refrigeration, automotive applications, and large-tonnage chillers.[8] Another highly relevant pure refrigerant is R-32 (difluoromethane), which is utilized extensively in high-efficiency mini-split systems and possesses excellent thermal conductivity and specific heat properties.[8] Azeotropic and near-azeotropic blends, like R-410A (a perfectly balanced 50/50 blend of R-32 and R-125), behave so similarly to pure refrigerants that their microscopic temperature variance during phase change is generally negligible in routine field diagnostics.[11] Similarly, R-404A is a near-azeotropic blend consisting of 44% R-125, 52%

R-143a, and 4% R-134a, widely deployed in low-temperature commercial refrigeration applications like walk-in freezers and supermarket display cases.[12]

As the industry aggressively transitions to next-generation low-GWP alternatives to meet environmental targets, zeotropic blends like R-454B have become prevalent.[8] R-454B is the primary A2L (mildly flammable) drop-in replacement candidate for R-410A in newly manufactured North American equipment, boasting a significantly reduced GWP of 466.[7]

R-454B is a zeotropic blend composed of 68.9% R-32 and 31.1% R-1234yf.[14] Because the constituent chemicals in a zeotropic blend have distinctly different boiling points, the mixture does not evaporate or condense at a single, constant temperature. Instead, the phase change occurs sequentially over a temperature range, a complex thermodynamic phenomenon known as temperature glide.[16]

## Temperature Glide and the Bubble and Dew Point Paradigm

For zeotropic blends like R-454B and older blends like R-407C, utilizing a standard pressure-temperature chart without understanding temperature glide will lead to catastrophic diagnostic errors and improper system charging.[16] The saturation charts for these specific refrigerants divide the saturation temperature into two distinct columns corresponding to the beginning and end of the phase change process: the Bubble Point and the Dew Point.[15]

The Bubble Point represents the specific temperature at which a saturated liquid first begins to boil—meaning the appearance of the very first bubble of vapor—at a specific system pressure. Below this exact temperature, the fluid exists completely as a subcooled liquid.[18] Conversely, the Dew Point represents the temperature at which a saturated vapor first begins to condense—marked by the appearance of the first drop of liquid dew—at a specific pressure. Above this temperature, the fluid is completely superheated vapor.[18]

When calculating advanced diagnostic metrics in the field, the correct saturation value must be meticulously cross-referenced based on the state of the fluid being measured. Subcooling is a measure of the liquid state occurring in the condenser. Therefore, it must be calculated using the Bubble Point value from the saturation table.[21] Subtracting the actual measured liquid line temperature from the bubble point temperature yields the mathematically correct subcooling value. Superheat, on the other hand, is a measure of the vapor state exiting the evaporator. Therefore, it must be calculated using the Dew Point value.[21] Subtracting the dew point temperature from the actual measured suction line temperature yields the correct superheat value. A common industry mnemonic to remember this critical distinction is "bubcool" for bubble-point subcooling, and "dewperheat" for dew-point superheating.[18]

While R-454B exhibits a relatively small temperature glide of approximately 1.5°F (0.8°C) compared to older zeotropic blends, this variance is mathematically significant enough to skew critical charging calculations and superheat targets, especially in cold-weather operation or when servicing highly tuned variable-speed inverter systems.[9] Using the incorrect column on a digital manifold or manual chart will result in an artificially high or low reading, prompting technicians to mistakenly add or recover refrigerant from a perfectly balanced system.[23]

To illustrate the vast differences in operating pressures across modern refrigerants, it is necessary to compare their saturation pressures at standard operating temperatures. At a typical evaporator saturation temperature of 40°F, R-134a operates at a low 35.0 psig.[24] R-404A operates at 85.4 psig.[25] R-454B (using the dew point for vapor) operates at 107.0 psig, while its bubble point is 112.0 psig.[24] R-410A operates at 118.0 psig, and R-32 operates at 118.8 psig.[10] At a typical condensing saturation temperature of 100°F, the differences become even more pronounced. R-134a condenses at 124.2 psig (interpolated), R-404A condenses at 234.6 psig, R-454B condenses at 290.9 psig (dew) to 300.8 psig (bubble), R-410A condenses at approximately 317.0 psig, and R-32 condenses at 325.7 psig.[10] These variations highlight why a universal diagnostic approach is impossible without specific pressure-temperature data matching the fluid in the circuit.

| Refrigerant | Type | Flammability (ASHRAE) | Saturation Pressure at 40°F (psig) | Saturation Pressure at 100°F (psig) |
|---|---|---|---|---|
| **R-134a** | Pure HFC | A1 (Non-flammable) | 35.0 | 124.2 |
| **R-404A** | Near-Azeotropic Blend | A1 (Non-flammable) | 85.4 | 234.6 |
| **R-410A** | Near-Azeotropic Blend | A1 (Non-flammable) | 118.0 | 317.0 (Approximate) |
| **R-32** | Pure HFC | A2L (Mildly Flammable) | 118.8 | 325.7 |
| **R-454B** | Zeotropic Blend (1.5°F Glide) | A2L (Mildly Flammable) | 107.0 (Dew) / 112.0 (Bubble) | 290.9 (Dew) / 300.8 (Bubble) |

# The Pressure-Enthalpy (P-H) Diagram as a Diagnostic Map

A Pressure-Enthalpy (P-H) diagram, occasionally referred to as a Mollier diagram depending on regional engineering nomenclature, graphically maps the thermodynamic properties of a refrigerant fluid across all physical states.[1] The vertical axis of the chart represents absolute pressure, typically expressed in pounds per square inch absolute (psia) or bar. Technicians must remember that gauge pressure (psig) requires the addition of atmospheric pressure (14.7 psi at sea level) to align with the chart's absolute values.[2] The horizontal axis represents specific enthalpy, which is a measure of the total heat content of the refrigerant, expressed in British Thermal Units per pound (Btu/lb) or kilojoules per kilogram (kJ/kg).[2]

The most prominent architectural feature of the P-H diagram is the saturation curve, universally referred to as the "vapor dome." The left side of this encompassing dome represents the saturated liquid line, where the refrigerant quality ( $x$ ) is mathematically equal to 0.[30] The right side of the dome represents the saturated vapor line, where the refrigerant quality ( $x$ ) is equal to 1.[30] The exact apex where these two continuous lines meet is defined as the critical point. If the fluid is pushed above this critical temperature and pressure threshold, the thermodynamic distinction between liquid and vapor ceases to exist, and the fluid enters a supercritical state.[29]

The diagram is interlaced with several families of complex isoclines, or lines of constant value. Isobars run horizontally, indicating lines of constant pressure.[2] Isenthalps run perfectly vertically, indicating lines of constant total heat content.[2] Isotherms represent lines of constant temperature; in the subcooled liquid region on the far left, they drop almost vertically, while inside the two-phase dome, they run horizontally parallel to the isobars. Upon exiting the dome into the superheated region on the right, isotherms curve sharply downwards.[29] Isentropes, representing lines of constant entropy, curve aggressively upward and to the right in the superheated region, dictating the theoretical thermodynamic path of vapor compression.[29] Finally, isochores denote lines of constant specific volume, which are instrumental for calculating the exact volumetric efficiency of the compressor under varying load conditions.[29]

By plotting the cycle on this mathematical grid, the refrigerant can be tracked through four distinct states, or diagnostic "quadrants." The subcooled liquid region resides to the left of the dome. Here, the refrigerant is a 100% dense liquid, and its position strictly dictates the total enthalpy entering the expansion device.[29] The two-phase mixture region is located entirely inside the dome. This zone represents the latent heat transfer areas of the evaporator and condenser, where the quality of the refrigerant ( $x$ ) denotes the mass fraction of vapor in the boiling or condensing mixture.[30] The superheated vapor region is located to the right of the dome. This maps the state of the gas entering and exiting the compressor, where the horizontal distance measured from the saturated vapor line represents the sensible heat gain.[33] Finally, the supercritical region exists above the critical point apex. While standard direct-expansion HVAC equipment avoids this zone, transcritical carbon dioxide (R-744) refrigeration systems operate extensively in this upper echelon, utilizing gas coolers to reject heat rather than traditional condensing coils.[9]

## Plotting the Actual Operating Cycle

Plotting field measurements onto a P-H diagram exposes exactly where an actual system deviates from its theoretical design parameters. The actual operating cycle deviates from the ideal cycle due to pressure drops in the suction and liquid lines, superheating of the vapor at the compressor inlet, and isentropic inefficiencies during the physical compression process. The procedure demands five critical field measurements taken simultaneously at steady state: high-side pressure, low-side pressure, condenser outlet temperature, evaporator outlet temperature, and compressor inlet temperature.[2]

The evaluation begins with the evaporation process. The refrigerant enters the evaporator coil as a cold, low-pressure, partial liquid-vapor mixture, having just passed through the expansion device.[36] The evaporation process is plotted as a line moving rightward along the low-pressure isobar.[31] In a theoretical, ideal cycle, this line is perfectly horizontal. In reality, a slight downward slope is plotted to mathematically account for the friction-induced pressure drop through the evaporator coil tubes and the return suction line.[38] The line eventually crosses the saturated vapor curve and terminates in the superheated vapor region at the exact measured compressor inlet temperature.

The compression process follows. The theoretical, perfect compression process is isentropic, meaning it operates at constant entropy. This is plotted by following the curved isentropic lines upward from the suction pressure to the discharge pressure.[35] However, real-world mechanical compression is never perfectly efficient. Heat is generated by friction, and motor windings impart thermal energy directly into the refrigerant gas. Consequently, the actual compression path diverges to the right of the constant entropy curve, generating excess enthalpy. The isentropic efficiency ($\eta_s$) of the compressor can be calculated using the specific enthalpies ($h$) extracted directly from the plotted P-H diagram. The formula is expressed as the theoretical isentropic discharge enthalpy minus the suction enthalpy, divided by the true measured actual discharge enthalpy minus the suction enthalpy.[40] A drastically rightward-leaning compression curve on the diagram definitively proves severe mechanical inefficiency, motor overheating, or internal valve leakage bypassing hot gas back into the suction chamber.[43]

Following compression is the condensation process. The highly superheated discharge gas enters the condenser, represented by a leftward movement along the high-pressure isobar. The gas must first shed its sensible heat, desuperheating until its temperature drops to strike the saturated vapor line at the dew point. It then traverses horizontally across the interior of the two-phase dome, representing latent condensation, until it hits the saturated liquid line at the bubble point.[33] Finally, the fluid continues its leftward trajectory past the dome boundary and into the subcooled liquid region, terminating exactly at the measured liquid line temperature.[37]

The final step is the expansion process. Expansion through a metering device is fundamentally an isenthalpic process, meaning the total heat content remains constant.[32] Therefore, it is plotted as a perfectly straight vertical line dropping precipitously from the high-pressure subcooled liquid point down to the low suction pressure line.[36] Because expansion is a violent throttling process that performs no useful thermodynamic work, the enthalpy entering the valve equals the enthalpy exiting the valve.[38]

By calculating the horizontal distance inside the two-phase dome along the bottom evaporation line, the analyst determines the Net Refrigeration Effect (NRE) in Btu/lb. Multiplying this enthalpy difference by the total mass flow rate of the refrigerant yields the actual, real-time total cooling capacity of the system, a metric that cannot be deduced from pressure gauges alone.[2]

# Maintenance-Specific Diagnostic Signatures on the P-H Diagram

When physical, mechanical, or environmental faults occur within the refrigeration circuit, the geometric shape of the plotted P-H cycle morphs in highly predictable ways. Recognizing these thermodynamic signatures enables rapid and mathematically provable root-cause isolation.

## Impairments in Heat Transfer: Condensers and Evaporators

A fouled or dirty condenser coil, restricted outdoor airflow from a failing fan motor, or the recirculation of hot discharge air severely impacts the system's ability to reject heat to the ambient environment. A compromised condenser cannot efficiently reject the Total Heat of Rejection (THOR), which consists of both the heat absorbed by the evaporator and the heat of compression added by the motor.[2] To compensate for the reduced heat transfer coefficient, the physical laws of thermodynamics dictate that the system must force the temperature differential between the refrigerant and ambient air wider. This results in a severe, rapid elevation of the condensing pressure.[44] On the P-H diagram, the entire top horizontal line of the cycle shifts upward on the vertical axis. The compression ratio skyrockets, and because the compression curve follows the divergent entropy lines upward and to the right, the final discharge temperature is violently elevated.[38] Depending on the specific metering device employed, subcooling may appear deceptively normal or slightly high as liquid backs up in the choked coil.[48]

Conversely, a dirty evaporator coil, restricted indoor airflow from a plugged media filter, or a failing indoor blower motor prevents the refrigerant from absorbing the requisite heat to boil at design conditions. On the P-H diagram, the bottom horizontal evaporation line drops downward on the vertical pressure axis, resulting in abnormally low suction pressure. Because the liquid refrigerant fails to fully vaporize early in the coil passes, the saturation temperature falls rapidly, often dropping below the 32°F (0°C) threshold, which results in atmospheric moisture freezing and accumulating as frost on the coil surface.[49] As a result of the reduced heat absorption, superheat plummets, extending the end of the evaporation line dangerously close to the vapor dome boundary and creating a severe risk of liquid floodback into the compressor crankcase.[48]

## Charge Anomalies: Undercharge and Overcharge

A system undercharge indicates that the total refrigerant volume is insufficient to satisfy the internal volumetric demands of the circuit.[50] The P-H signature is highly distinct. The condenser is starved of adequate mass, significantly reducing the necessary liquid seal at the base of the coil. This drives subcooling near zero, meaning the right-side starting point of the vertical expansion line shifts inward toward the vapor dome boundary. The evaporator is similarly starved; the meager liquid volume that reaches the expansion device boils off almost immediately upon entering the coil. The remaining massive surface area of the coil acts purely to absorb excess sensible heat, sending highly superheated gas back to the compressor.[50] On the diagram, the bottom line extends far to the right, deep into the superheated region.[6] Due to the lack of mass, both the overall high-side and low-side pressures drop below nominal parameters.[11]

A system overcharge creates the opposite problem; excess refrigerant mass physically floods the internal volumes of the heat exchangers.[6] Liquid backs up deep into the condenser tubes, drastically reducing its effective condensing surface area. This physical flooding acts thermodynamically identical to a dirty coil, pushing the head pressure rapidly upward.[47] Subcooling becomes excessively high, shifting the top of the vertical expansion line deeply to the left into the subcooled region. Consequently, the metering device is forced to overfeed the evaporator with liquid. Because the evaporator surface area is overwhelmed, the refrigerant barely has time to absorb latent heat before exiting the coil, leaving the vapor superheat near zero.[48] This condition is particularly dangerous as it mathematically guarantees liquid slugging if not immediately corrected.[50]

## The Impact of Non-Condensable Gases and Dalton's Law

Air, nitrogen, or other non-condensable gases (NCGs) introduced into the system through poor evacuation practices, degraded hoses, or inadequate vacuum pump performance remain in a gaseous state throughout the entire refrigeration cycle. Because they do not condense at normal operating pressures, they are swept through the discharge line and gather in the top passes of the condenser or within the upper volume of the liquid receiver.[53]

The thermodynamic signature of NCGs is governed strictly by Dalton’s Law of Partial Pressures. This fundamental gas law dictates that the total pressure of a gas mixture is mathematically equal to the sum of the partial pressures of its individual components.[53] Consequently, the total pressure measured on the high-side gauge consists of the actual saturated pressure of the refrigerant plus the partial pressure of the trapped nitrogen or air. On the P-H diagram, the high-side pressure line is artificially elevated far beyond the saturation pressure dictated by the ambient temperature, simulating an extreme overcharge or dirty condenser.[53] However, unlike an overcharge, the presence of NCGs is often accompanied by erratic, rapidly bouncing high-side gauge needles as unpredictable pockets of non-condensable gas pass violently through the compressor discharge reed valves.[28]

## Mechanical Flow Restrictions and Internal Bypassing

A restricted liquid line, often caused by a severely clogged filter-drier, a kinked copper line set, or internal debris lodging in the expansion valve screen, creates an unintended and highly detrimental throttling point in the high side of the system.[50] The P-H signature reveals a premature vertical pressure drop occurring on the diagram long before the fluid reaches the actual primary expansion device.[57] Because this pressure drop happens while the fluid is in a subcooled liquid state, a portion of the liquid flashes to vapor prematurely. This flashing absorbs latent heat from the surrounding fluid, creating a sensible and measurable temperature drop across the restricted component.[50] The primary expansion device is starved of liquid, leading to high superheat and low suction pressure, while the unmoving liquid backing up behind the restriction in the condenser causes high subcooling.[29]

If the thermal expansion valve (TXV) sensing bulb loses thermal contact with the suction line or its internal mechanical spring degrades, the valve's intricate proportional-integral-derivative (PID) mechanical balancing act fails. The P-H signature shows a system that never achieves steady state. On the diagram, the point of superheat and the horizontal suction pressure line continually oscillate back and forth. The valve erratically over-feeds the evaporator (dropping superheat), realizes its error, and then heavily under-feeds the evaporator (spiking superheat), a destructive cyclical process known as hunting.[59]

Internal mechanical bypassing represents a catastrophic failure of system compartmentalization. Compressor valve failure occurs when damaged suction or discharge reed valves, or worn internal scrolls, allow high-pressure discharge gas to bleed continuously back into the low-pressure suction cylinder during the intake stroke.[51] The system completely loses its volumetric pumping capacity. The suction pressure line rises abnormally high on the P-H chart, and the discharge pressure line drops abnormally low; the dynamic pressures constantly attempt to equalize while the motor is running.[15] The actual compression path on the diagram veers drastically to the right into the superheated zone due to the extreme entropy generation and localized heat buildup.[57]

A highly similar bypassing scenario occurs in heat pump applications if the internal teflon slide of the reversing valve degrades or leaks. Hot, high-pressure discharge gas mixes directly with cool, low-pressure suction gas returning to the compressor.[48] The P-H signature reveals that the suction gas temperature entering the compressor is significantly elevated. A physical temperature delta measurement greater than 3°F across the suction piping before and after the reversing valve confirms the internal leak.[65] This artificially high superheat drastically lowers the density of the gas entering the compressor, severely crippling the mass flow rate and destroying total system heating and cooling capacity.[66]

## The Superheat and Subcooling Cross-Reference Diagnostic Matrix

While plotting the full thermodynamic cycle on a P-H diagram provides a comprehensive engineering view, field technicians often utilize a highly effective, truncated version of thermodynamic analysis known as the four-quadrant diagnostic matrix. By measuring just two absolute values—system Superheat (SH) representing the evaporator and system Subcooling (SC) representing the condenser—and categorizing them as Normal, High, or Low relative to manufacturer specifications, the fundamental mechanism of the fault can be rapidly isolated from a wide field of possibilities.[67]

## Matriz de Diagnóstico Termodinâmico: Superaquecimento vs. Subarrefecimento

fecimento ➡
ecimento ⬇

| | BAIXO SUBARREFECIMENTO | ALTO SUBARREFECIMENTO |
|---|---|---|
| ALTO SUPERAQUECIMENTO | **Carga Baixa (Subcarga)**<br>O evaporador não tem fluido suficiente, fazendo com que o líquido ferva cedo na serpentina. Do lado de alta pressão, o condensador também carece de refrigerante para preencher a sua porção inferior. **Ação:** Verifique rigorosamente se existem fugas nas válvulas ou linhas antes de adicionar carga. | **Restrição**<br>Uma restrição na linha de líquido ou no dispositivo de expansão (TXV) retém o fluido refrigerante no lado de alta pressão enquanto priva o evaporador de fluxo. **Causas comuns:** Filtro secador entupido, linha de líquido dobrada ou TXV parcialmente bloqueada por contaminantes. |
| BAIXO SUPERAQUECIMENTO | **Problema no Compressor ou TXV**<br>O compressor pode apresentar válvulas fracas e bombeamento deficiente. Em alternativa, a TXV pode estar presa aberta, inundando o evaporador, enquanto o condensador não consegue gerar pressão suficiente para subarrefecer o líquido. Também pode ser indício de falha na válvula inversora. | **Sobrecarga**<br>Existe demasiado fluido refrigerante no sistema. O condensador fica inundado, o que aumenta drasticamente a pressão de alta. Simultaneamente, o evaporador é sobrealimentado, correndo o risco grave de retorno de líquido em fase líquida direto ao compressor. **Ação:** Recuperar o excesso de carga. |

O cruzamento do Superaquecimento do sistema (desempenho do evaporador) com o Subarrefecimento (desempenho do condensador) isola o domínio principal de falhas mecânicas ou relacionadas com a carga térmica.

Fontes de dados: Ferguson, Intrysys, Oxmaint, HVAC School

The definitive indications of the four diagnostic matrix combinations provide clear directives for intervention.

High Superheat combined with Low Subcooling dictates a definitive system undercharge. This remains the most common diagnostic pattern encountered in the field.[6] A lack of total physical refrigerant mass in the circuit dictates that the condenser cannot stack enough liquid at its base to establish proper subcooling targets.[50] Concurrently, the evaporator is actively starved of liquid; the meager refrigerant flashes to vapor immediately upon passing the metering device and absorbs excess sensible heat across the vast remaining coil surface, sending highly superheated, low-density gas to the compressor.[50] The diagnosis necessitates a rigorous electronic leak search, as closed hermetic systems do not consume refrigerant through normal operation.[6]

Low Superheat combined with High Subcooling dictates a definitive system overcharge. Excess mass physically fills the condenser tubes with dense liquid, backing up the head pressure and driving subcooling aggressively high above target ranges.[50] The metering device is physically forced by the high head pressure to overfeed the evaporator with liquid. Because the evaporator surface area is overwhelmed by the fluid volume, the refrigerant barely has time to absorb latent heat from the airflow before exiting the coil structure, leaving the superheat near zero.[48] This condition is critical, as it risks liquid droplets entering the compressor cylinders.[50]

High Superheat combined with High Subcooling isolates a severe liquid line restriction. This combination definitively isolates a mechanical flow issue, separating it conceptually from simple charge volume issues.[15] A physical restriction, such as a kinked copper line, plugged filter-drier, or a partially stuck-closed TXV, traps the bulk of the refrigerant mass in the high side. The condenser fills with backed-up liquid, driving subcooling high.[48] The restriction starves the downstream evaporator of vital flow, driving superheat excessively high.[50] The technician must verify this specific diagnosis by seeking an anomalous, localized temperature drop across liquid line components using a contact thermocouple.[50]

Low Superheat combined with Low Subcooling points to a catastrophic failure of system compartmentalization, generally internal bypassing or total metering failure. This rare but devastating signature occurs when the compressor suffers shattered internal valves and fails to establish a functional pressure differential, or when the TXV is stuck wide open, dumping hot liquid into the evaporator so fast that the condenser cannot properly subcool it and the evaporator cannot properly superheat it before it is drawn back to the suction port.[50] A internally bypassing reversing valve on a heat pump yields identical symptoms.[48]

# Inverter System Diagnostics: Variable-Speed Compressors and EEVs

The standard paradigm of HVAC diagnostics is fundamentally disrupted by the ubiquitous integration of Variable-Frequency Drive (VFD) driven inverter compressors and Electronic Expansion Valves (EEVs). While traditional single-stage systems operate at a fixed volumetric displacement, relying entirely on the ambient heat load to naturally dictate pressure variations, inverter systems are highly active, dynamic machines that mask underlying thermodynamic faults.

In an inverter system, the compressor speed dynamically modulates from roughly 30% to 120% of its nominal design capacity based on the real-time thermal load of the conditioned space, communicated via indoor temperature sensors.[6] Simultaneously, the EEV modulates its internal stepper motor via a localized PID loop driven by electronic thermistors to maintain a tightly controlled, pre-programmed target superheat.[49]

Because the EEV dynamically and continuously adjusts to the compressor's shifting mass flow variations, analyzing superheat alone under partial load conditions is entirely futile.[49] If a system is slightly undercharged, an intelligent EEV will simply step open wider to compensate, maintaining a perfect factory superheat target while quietly sacrificing subcooling and compressor motor cooling.[49] Conversely, if the system is moderately overcharged, the EEV will throttle back to choke the flow and prevent liquid floodback.[49] This active modulation creates a scenario where a system with an incorrect charge appears to have perfect low-side thermodynamic parameters until it is pushed to its absolute capacity limits during peak ambient conditions, at which point it fails.

To extract actionable thermodynamic data, these active dynamic variables must be artificially frozen. Field diagnostics on inverter systems absolutely mandate initiating the manufacturer-specific "Charge Mode," "Test Mode," or "System Verification Mode" through the proprietary thermostat interface or control board dip switches.[71] Initiating this mode forces the inverter drive logic to lock the compressor at a steady-state speed, typically 100% of its nominal capacity, and fixes the EEV at a specific step-position or allows it to target a fixed superheat without interference from shifting compressor mass flow.[75] Only when the system is forcefully stabilized in this manner can standard P-H charting and the SH/SC diagnostic matrix be reliably applied.[72] If a technician attempts to evaluate pressures while the system is ramping up or modulating down, the fluctuating mass flow will render the pressure-enthalpy calculations mathematically invalid and practically useless.

# Practical Field Measurement Procedures and Instrumentation

The analytical accuracy of advanced thermodynamic diagnostics is entirely bound to the physical precision of the field measurements. In high-efficiency equipment, a deviation of just 2°F to 3°F in temperature readings can drastically skew superheat and subcooling calculations, pushing the diagnostic mapping out of the optimal operating envelope and leading to an incorrect charge assessment. This minor error can cause up to a 15% reduction in overall system energy efficiency and increase the risk of compressor failure by over 10%.[5]

The industry shift from traditional analog mechanical gauges, which rely on expanding Bourdon tubes, to sophisticated digital manifolds is arguably the most critical upgrade in modern diagnostic instrumentation.[78] Digital manifolds, such as the widely adopted Fieldpiece SMAN series, offer massive advantages over analog counterparts.[80] Digital pressure transducers provide pinpoint accuracy and high resolution without the parallax reading error inherent to analog dials.[78] More importantly, digital manifolds contain internal, constantly updated NIST-grade pressure-temperature databases for numerous modern refrigerants, including R-32, R-410A, and R-454B.[79] For zeotropic blends like R-454B and R-407C, the manifold's microprocessor automatically segregates the Bubble and Dew point saturation curves. It calculates superheat strictly off the dew point and subcooling strictly off the bubble point in real-time, eliminating the severe calculation errors often made when technicians manually cross-reference a physical paper P-T chart.[18]

| Feature / Capability | Analog Manifold Gauges | Digital Manifold Gauges |
|---|---|---|
| **Measurement Mechanism** | Mechanical Bourdon Tube | Digital Pressure Transducers |
| **Saturation Temperature** | Requires manual lookup on P-T charts | Automatic, real-time NIST-grade database |
| **Zeotropic Blend Handling** | High risk of confusing bubble/dew points | Automatically segregates bubble and dew points |
| **Superheat/Subcooling Math** | Manual calculation required | Continuous automatic calculation |
| **Data Logging / Connectivity** | None | Bluetooth integration, psychrometric pairing |
| **Durability vs. Precision** | Highly durable, subject to parallax error | Sensitive electronics, absolute precision |

To accurately plot the endpoints on the P-H diagram, temperature measurements must perfectly capture the temperature of the refrigerant flowing inside the pipe, insulated entirely from the ambient air surrounding it.[52] Strap-on thermistors or K-type thermocouple pipe clamps must be securely fastened to completely clean, unoxidized sections of the copper tubing. They should be mechanically insulated from ambient air currents, direct sunlight, or rain during the measurement process.[83] Modern thermistors and thermocouples are subject to gradual thermal drift over time due to repeated thermal shock and environmental degradation. Annual calibration of these sensors to ISO/IEC 17025 standards is highly recommended to maintain the tight tolerances required for inverter system diagnostics.[5]

The physical laws of thermodynamics require dedicated time for heat transfer to reach equilibrium across massive metal coils and varying fluid volumes. After initial system startup, or after any adjustment to the refrigerant charge via addition or recovery, the technician must allow a strict minimum stabilization period before recording data.[83] Research indicates that while temperatures like suction superheat and liquid subcooling may appear generally stable after 4 to 5 minutes of run time, complex variables like compressor discharge temperatures and high-side mass flow dynamics may require up to 10 to 15 minutes of continuous, uninterrupted operation to achieve true steady-state equilibrium.[72] Taking measurements prematurely will result in plotting a ghost cycle on the P-H diagram that does not reflect actual, sustained operating conditions, leading to false diagnostic conclusions.

# Applied Diagnostic Case Studies in Thermodynamic Analysis

To synthesize the preceding thermodynamic theories, fluid dynamics, and instrumentation protocols, the following worked examples demonstrate the rigorous step-by-step diagnostic reasoning required in the field when faced with actual data.

## Case 1: R-454B Split System in Cooling Mode

A technician is dispatched to evaluate a newly installed 3-Ton R-454B direct-expansion split system equipped with a traditional thermostatic expansion valve (TXV).[16] The system is failing to satisfy the space temperature. The technician connects a digital manifold, applies properly insulated temperature clamps, and allows the system to stabilize for 15 minutes to reach steady state.

The recorded field measurements are:

- Suction Pressure: 95 psig
- Discharge Pressure: 430 psig
- Suction Line Temperature: 65°F
- Liquid Line Temperature: 90°F

The thermodynamic analysis begins with identifying the refrigerant properties. R-454B is an A2L zeotropic blend with temperature glide.[15] The technician must utilize the R-454B pressure-temperature data, strictly observing the division between Dew and Bubble points.[24] To analyze the low side and determine superheat, the technician references the Dew Point for R-454B at 95 psig. Interpolating the specific P-T data, 95 psig corresponds to a saturated vapor Dew Point of approximately 33°F.[24] The superheat is mathematically derived by subtracting the Dew Point (33°F) from the measured Suction Line Temperature (65°F), resulting in an extreme 32°F Superheat. This is exceptionally high, as normal TXV targets are typically 10°F to 15°F.[16]

To analyze the high side and determine subcooling, the technician references the Bubble Point for R-454B at 430 psig. The chart dictates a saturated liquid Bubble Point of approximately 126°F.[24] The subcooling is derived by subtracting the measured Liquid Line Temperature (90°F) from the Bubble Point (126°F), resulting in an extreme 36°F Subcooling. This is exceptionally high, as normal targets are typically 8°F to 12°F.[16]

Applying the diagnostic matrix, the technician is presented with High Superheat (32°F) and High Subcooling (36°F). Referencing the established thermodynamic logic, the simultaneous presence of high superheat and high subcooling definitively proves a severe liquid line restriction.[15] The restriction, likely a clogged internal filter-drier from poor brazing practices, is creating a massive pressure drop mid-line, stacking unmoving liquid in the condenser (causing the 36°F SC) and completely starving the downstream evaporator of flow (causing the 32°F SH).

## Case 2: R-410A Commercial Package Unit

A technician evaluates an aging 5-Ton R-410A commercial package unit utilizing a fixed-speed

scroll compressor. The building manager reports the air from the supply registers is tepid despite the compressor running continuously.
The recorded field measurements are:

- Suction Pressure: 173 psig
- Discharge Pressure: 238 psig
- Suction Line Temperature: 68°F
- Compressor Amp Draw: 40% below Rated Load Amps (RLA)

The thermodynamic analysis begins by analyzing the absolute pressures against the ambient baseline. R-410A operates at distinctly higher baseline pressures than legacy systems. A suction pressure of 173 psig equates to a saturation temperature of approximately 60°F. A discharge pressure of 238 psig equates to a saturation temperature of 80°F.[11] A 60°F evaporating temperature is abnormally high for a space conditioning application, where typical pressures hover around 118 psig for a 40°F coil.[11] Conversely, an 80°F condensing temperature is impossibly low if the outdoor ambient environment is 85°F or warmer.[11]
The primary indicator is the compression ratio. The ratio of absolute discharge pressure to absolute suction pressure is extremely low. The high suction and low discharge indicate the compressor is completely failing to build a functional pressure differential between the high and low sides of the system.[11] Coupled with the low electrical amp draw indicating the motor is doing no actual compression work, this pressure profile is the classic thermodynamic signature of catastrophic compressor valve failure.[15] High-pressure discharge gas is bypassing internally, bleeding continuously back into the suction chamber. A secondary mechanical check to attempt pumping the system down into a vacuum will fail, confirming shattered internal valves.[63] The compressor must be replaced.

## Case 3: R-404A Low-Temperature Refrigeration System

A technician is troubleshooting a commercial walk-in freezer utilizing R-404A. The compressor is frequently tripping on high-head pressure limits during peak afternoon ambient conditions.
The recorded field measurements are:

- Discharge Pressure: 450 psig
- Liquid Line Temperature: 95°F
- Ambient Outdoor Temperature: 80°F

The technician begins by evaluating the condensing parameters. At 450 psig, R-404A has a saturated condensing temperature of approximately 130°F.[13] The calculated subcooling is the saturation temperature (130°F) minus the liquid line temperature (95°F), resulting in an extreme 35°F of subcooling. The temperature differential between the saturated condensing temperature (130°F) and the ambient outdoor air (80°F) is a massive 50°F split, far above the expected 15°F to 20°F split for an air-cooled condenser.
The technician must differentiate between a severe overcharge and the presence of non-condensable gases (NCGs). Both scenarios drastically elevate head pressure. An extreme overcharge physically backs liquid up into the condenser, reducing active surface area and causing massive subcooling (35°F). However, if NCGs were present, Dalton's Law of Partial Pressures would dictate that the 450 psig reading is a combination of refrigerant saturation

pressure and trapped air pressure. If the system was heavily contaminated with air but correctly charged with liquid, the liquid line temperature would typically remain closer to the true, un-elevated saturation temperature, resulting in an artificially calculated high subcooling number but often accompanied by erratic, wildly bouncing gauge needles as air pockets hit the compressor discharge valves.[28] Because the gauge needles are completely steady and the subcooling is massively elevated with a solid column of liquid, the thermodynamic diagnosis points to a gross overcharge. The technician must recover the excess R-404A charge until the subcooling drops into the manufacturer's specified range, at which point the head pressure will normalize.

Through these rigorous thermodynamic applications, technicians elevate their diagnostic capabilities, ensuring peak efficiency, preventing catastrophic mechanical failures, and maintaining the exact performance standards required by modern, highly sophisticated climate control systems.

## Referências citadas


1. Pressure-Enthalpy Diagrams: Aturation Ubcooling AND Uperheat | PDF - Scribd, acessado em junho 13, 2026, https://www.scribd.com/document/46336659/HVAC-Ch09
2. Pressure Enthalpy Without Tears w/ Eugene Silberstein - YouTube, acessado em junho 13, 2026, https://www.youtube.com/watch?v=JgwaPyjMzk4
3. Simplified Guide To Superheat and Subcool : r/HVAC - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/HVAC/comments/1hgd7la/simplified_guide_to_superheat_and_subcool/
4. What Do Superheat And Subcooling Mean In HVAC?, acessado em junho 13, 2026, https://eticampus.edu/hvacr-program/hvac-training/what-do-superheat-and-subcooling-mean-in-hvac/
5. Superheat & Subcooling Calibration: Precision Thermal Management, acessado em junho 13, 2026, https://techmaster.us/superheat-subcooling-calibration-precision-thermal-management/
6. HVAC Superheat & Subcooling: Pro Diagnostic Guide - Ferguson, acessado em junho 13, 2026, https://www.ferguson.com/content/ideas-and-learning-center/trade-talk/hvac-superheat-subcooling/
7. Refrigerant Transition Update The Impact of Refrigerant Choice on the Design Sustainability of Your Projects - Trane, acessado em junho 13, 2026, https://www.trane.com/content/dam/Trane/Commercial/north-america/about-us/newsroom/events/2023-engineer-summit/2023-EngSummit-Refrigerants-Update-FINAL.pdf
8. Refrigerant P-T Chart Guide - ToolGrit, acessado em junho 13, 2026, https://www.toolgrit.com/guides/refrigerant-pt-guide

9. R-407A Pros, Cons & Comparisons to Other Refrigerants | The Super Blog, acessado em junho 13, 2026, https://www.superradiatorcoils.com/blog/r-407a-pros-cons-comparisons-r-404a-r-22
10. R-32 REFRIGERANT: GET THE FACTS, acessado em junho 13, 2026, https://apps.goodmanmfg.com/brochures/files/68755da8f3a7bCB-R32-GETFACTS_06-25_PROOF1.pdf
11. R-410A Pressure Chart for HVAC - Royal Refrigerants, acessado em junho 13, 2026, https://royalrefrigerants.com/blogs/news/r-410a-pressure-chart-for-hvac
12. The Ultimate Guide to the R404 Pressure-Temperature Chart - American Mortuary Coolers, acessado em junho 13, 2026, https://www.mymortuarycooler.com/blogs/news/the-ultimate-guide-to-the-r404-pressure-temperature-chart
13. R-404A Pressure Temperature Chart, acessado em junho 13, 2026, https://www.hudsontech.com/pdfs/pt-charts/R-404A-Pressure-Temperature-Chart.pdf
14. R-454B-Pressure-Temperature-Chart.pdf, acessado em junho 13, 2026, https://www.hudsontech.com/pdfs/pt-charts/R-454B-Pressure-Temperature-Chart.pdf
15. How to Read a Refrigerant Pressure Chart: A2L, HFC & HFO Subcool/Superheat Guide, acessado em junho 13, 2026, https://oxmaint.com/industries/hvac/how-to-read-refrigerant-pressure-chart-a2l-hfc-hfo-subcool-superheat
16. R-454B vs R-410A Pressures: A Side-by-Side PT Chart Comparison - AC Direct, acessado em junho 13, 2026, https://www.acdirect.com/blog/r454b-vs-r410a-pressures/
17. acessado em junho 13, 2026, https://hvacknowitall.com/blog/r-454b-cold-weather-charging-why-subcooling-fails-below-55f#:~:text=R%2D454B's%20near%2Dazeotropic%20blend,%C2%B0F%20glide%20is%20manageable.
18. Glide, Dew Point, Bubble Point, PT Charts and the Refrigerant Slider App - YouTube, acessado em junho 13, 2026, https://www.youtube.com/watch?v=4B11Jkk1W-8
19. How to read a pressure temperature chart for super-heat and sub-cooling calculations, acessado em junho 13, 2026, https://www.se.com/be/nl/faqs/FA168444/
20. Technical data sheet R-454B | Gas Servei, acessado em junho 13, 2026, https://gas-servei.com/wp-content/uploads/2024/09/Technical-data-sheet-R-454B-Gas-Servei-2023.pdf
21. Bubcool and Dewperheat (Bubble and Dew Point explained ..., acessado em junho 13, 2026, http://www.hvacrschool.com/videos/bubcool-and-dewperheat-bubble-and-dew-point-explained/
22. The Definitive Guide to Temperature Glide, Part 2 | ACHR News, acessado em junho 13, 2026,

https://www.achrnews.com/articles/147494-the-definitive-guide-to-temperature-glide-part-2
23. How measureQuick Calculates Superheat & Subcooling, acessado em junho 13, 2026, https://support.measurequick.com/portal/en/kb/articles/how-measurequick-calculates-superheat-subcooling
24. Pressure Temperature Chart - Arkema Forane, acessado em junho 13, 2026, https://forane.arkema.com/files/live/sites/shared_arkema/files/downloads/fluorochemicals/Forane%20Ref%20PT%20Regular%20Chart%20EN
25. The Complete Guide to 404A Pressure-Temp Charts - American Mortuary Coolers, acessado em junho 13, 2026, https://www.mymortuarycooler.com/blogs/news/the-complete-guide-to-404a-pressure-temp-charts-1
26. R-134a Temperature - Pressure Chart - Carrier, acessado em junho 13, 2026, https://www.transcentral.carrier.com/CPGTechPubs/toolbox/temp-pressure-chart_r134a.html
27. Forane 134a Pressure Temperature Chart, acessado em junho 13, 2026, https://forane.arkema.com/files/live/sites/shared_arkema/files/downloads/products/fluorochemicals/forane-134a-pressure-temperature-chart.pdf
28. Engineering Terminology - Intech Bearing Inc., acessado em junho 13, 2026, https://intechbearing.com/pages/engineering-terms
29. MOLLIER Diagrams in Refrigeration and airconditioning technolology | PPTX - Slideshare, acessado em junho 13, 2026, https://www.slideshare.net/slideshow/mollier-diagrams-in-refrigeration-and-airconditioning-technolology/271480924
30. Thermodynamics Basics: Gas Dynamics, Psychrometric Analysis, Refrigeration Cycle and HVAC Systems - CEDengineering.com, acessado em junho 13, 2026, https://www.cedengineering.com/userfiles/M08-006%20-%20Thermodynamics%20Basics%20-%20Gas%20Dynamics,%20Psychrometric%20Analysis,%20Refrigeration%20Cycle%20and%20HVAC%20Systems%20-%20US.pdf
31. 2. How to draw a refrigeration cycle on a ph diagram (Question), acessado em junho 13, 2026, https://www.jsrae.or.jp/jsrae/stady/Eng%20saikuru.htm
32. The Pressure-Enthalpy Diagram | Part 1: Theory - YouTube, acessado em junho 13, 2026, https://www.youtube.com/watch?v=qDY-NIyXMzo
33. Pressure / Enthalpy Diagram Example - HVAC School, acessado em junho 13, 2026, http://www.hvacrschool.com/pressure-enthalpy-diagram-example/
34. Pressure – enthalpy diagram for R744, R404A and R290 refrigerants. - ResearchGate, acessado em junho 13, 2026, https://www.researchgate.net/figure/Pressure-enthalpy-diagram-for-R744-R404A-and-R290-refrigerants_fig2_237902675
35. Pressure Enthalpy Without Tears w/ Eugene Silberstein - HVAC School, acessado em junho 13, 2026, http://www.hvacrschool.com/videos/pressure-enthalpy-without-tears-w-eugene-silberstein/
36. Pressure Enthalpy Diagram - Engineering Pro Guides, acessado em junho 13,

2026, https://www.engproguides.com/pressure-enthalpy-diagram.html
37. 2.2.2 How to draw actual operating state on P-h Chart, acessado em junho 13, 2026, https://uomus.edu.iq/img/lectures21/MUCLecture_2022_81813315.pdf
38. Vapor Compression Cycle: Diagram, COP & Examples - Turn2engineering, acessado em junho 13, 2026, https://turn2engineering.com/mechanical-engineering/thermodynamics/vapor-compression-cycle
39. Case – 1 The Basics of Refrigeration Cycle PH Diagram Analysis & Refrigerant Flow Diagram | TS Wan's blog, acessado em junho 13, 2026, https://tswanblog.com/wp-content/uploads/2013/03/case-1-the-basics-of-refrigeration-cycle-p-h-diagram-analysis.pdf
40. Experimental Evaluation of a Miniature Rotary Compressor for Application in Electronics Cooling - Purdue e-Pubs, acessado em junho 13, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2928&context=icec
41. Identification of Key Parameters and Construction of Empirical Formulas for Isentropic and Volumetric Efficiency of High-Temperature Heat Pumps Based on XGBoost-MLR Algorithm - MDPI, acessado em junho 13, 2026, https://www.mdpi.com/1996-1073/18/16/4454
42. Compressor Calculations: Rigorous Using Equation of State vs Shortcut Method, acessado em junho 13, 2026, http://www.jmcampbell.com/tip-of-the-month/2011/11/compressor-calculations-rigorous-using-equation-of-state-vs-shortcut-method/
43. CYCLE_D-HX: NIST Vapor Compression Cycle Model Accounting for Refrigerant Thermodynamic and Transport Properties - NIST Technical Series Publications, acessado em junho 13, 2026, https://nvlpubs.nist.gov/nistpubs/TechnicalNotes/NIST.TN.2134.pdf
44. Liebert® MCV Installer/User Guide - Vertiv, acessado em junho 13, 2026, https://www.vertiv.com/48ea40/globalassets/products/thermal-management/liebert-mcv-installer-user-guide-sl-19539.pdf
45. Vertiv™ CoolPhase Condenser User Guide, acessado em junho 13, 2026, https://www.vertiv.com/498fe6/globalassets/shared/vertiv-coolphase-condenser-user-guide-sl-80173.pdf
46. US20070089434A1 - Monitoring refrigerant in a refrigeration system - Google Patents, acessado em junho 13, 2026, https://patents.google.com/patent/US20070089434A1/en
47. DOCUMENT RESUME Refrigeration and Air Conditioning Equipment, 11-9. Military Curriculum Materials for Vocational and Technical E - ERIC, acessado em junho 13, 2026, https://files.eric.ed.gov/fulltext/ED226240.pdf
48. The "5 Pillars" of Residential A/C Refrigerant Circuit Diagnosis - HVAC School, acessado em junho 13, 2026, http://www.hvacrschool.com/the-5-pillars-of-residential-ac-refrigerant-circuit-diagnosis/
49. Checking the Charge of a Mini-Split Unit! - AC Service Tech, acessado em junho 13, 2026, https://www.acservicetech.com/post/checking-the-charge-of-a-mini-split-unit

50. Superheat & Subcooling Calculator 2026 | Free HVAC Refrigerant Tool | Intry, acessado em junho 13, 2026, https://www.intrysys.com/tools/hvac/superheat-subcooling-calculator
51. Maintenance Checklist: Keeping a 4-Ton ... - The Furnace Outlet, acessado em junho 13, 2026, https://thefurnaceoutlet.com/blogs/hvac-tips/maintenance-checklist-keeping-a-4-ton-commercial-system-alive-for-20-years
52. A/C Diagnosis: Superheat & Subcooling Guide | PDF | Mechanical Engineering - Scribd, acessado em junho 13, 2026, https://www.scribd.com/document/519024040/Superheat-a-Sub-Cooling-Document
53. Diagnosing Lack of Compressor Capacity | PDF | Air Conditioning - Scribd, acessado em junho 13, 2026, https://www.scribd.com/document/244946800/Notes-Kotza-139-169
54. PURDUE UNIVERSITY - Howard's ups and downs, acessado em junho 13, 2026, https://howardcheung.github.io/raw_files/Final_dissertation_edit.pdf
55. WCAP-16996-NP, Revision 0, "Realistic LOCA Evaluation Methodology Applied to the Full Spectrum of Break Sizes (FULL SPECTRU - Nuclear Regulatory Commission, acessado em junho 13, 2026, https://www.nrc.gov/docs/ML1036/ML103610232.pdf
56. How do you diagnose non condensables? : r/HVAC - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/HVAC/comments/qs0r6u/how_do_you_diagnose_non_condensables/
57. Inverse Modeling to Simulate Fault Impacts for Vapor Compression Equipment Part 2: System Modeling and Validation - Purdue e-Pubs, acessado em junho 13, 2026, https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=2166&context=iracc
58. Evaluating the Effects of Common Faults on a Residential Split System - | ETCC, acessado em junho 13, 2026, https://www.etcc-ca.com/sites/default/files/reports/HT.11.SCE_.007%20Faults%20on%20a%20Residential%20Split%20System_Final.pdf
59. Carrier TXV Expansion Valve vs EEV: What HVAC Pros Should Know-PartsHnC, acessado em junho 13, 2026, https://www.partshnc.com/blog/carrier-txv-expansion-valve-vs-eev-what-hvac-pros-should-know
60. Parameters and setup - Arsystem, acessado em junho 13, 2026, https://www.arsystem.pl/carel/dokumentacje2003/02%20GB/02%20Prezentacje.%20szkolenia/+0709GB005.pdf
61. A Guide to the Refrigeration Expansion Valve - Putnam Mechanical, acessado em junho 13, 2026, https://www.pmechanicalfl.com/post/a-guide-to-the-refrigeration-expansion-valve
62. HVAC Compressor Failure: 9 Warning Signs and What Each One Means - Oxmaint, acessado em junho 13, 2026, https://oxmaint.com/industries/hvac/hvac-compressor-failure-warning-signs-diagnostics-troubleshooting

63. Bad compressor? : r/HVAC - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/HVAC/comments/1kqcc0y/bad_compressor/
64. T-s diagram for liquid line restriction scenarios. - ResearchGate, acessado em junho 13, 2026, https://www.researchgate.net/figure/T-s-diagram-for-liquid-line-restriction-scenarios_fig10_271910048
65. 4 Signs Your Reversing Valve is Stuck in Cooling Mode | Heating & Cooling | AC Repair, acessado em junho 13, 2026, https://qrc123.com/heat-pump-reversing-valve-repair/
66. Heat Pump Reversing Valves Explained: How They Work in HVAC Systems, acessado em junho 13, 2026, https://hvacknowitall.com/blog/heat-pump-reversing-valves-explained-how-they-work-in-hvac-systems
67. Superheat and Subcooling for Refrigerant Charge Diagnostics: PT, acessado em junho 13, 2026, https://calcengineer.com/blog/superheat-subcooling-refrigerant-charge-diagnostics/
68. HVACR Troubleshooting Fundamentals - Technical Training Associates, acessado em junho 13, 2026, https://techtrainassoc.com/wp-content/uploads/2018/05/ttahtf_lookinside.pdf
69. Introduction to Superheat and Subcooling Diagnostics | Contracting Business, acessado em junho 13, 2026, https://www.contractingbusiness.com/service/article/21247583/introduction-to-superheat-and-subcooling-diagnostics
70. VARIABLE SPEED R-410A AIR CONDITIONING OUTDOOR UNITS - Fujitsu General, acessado em junho 13, 2026, https://www.fujitsugeneral.com/us/resources/pdf/support/downloads/installation-instructions/pdf-92-104921-12-03-fo-20c.pdf
71. Service Facts Variable Speed ComfortLink II Heat Pumps and Air Conditioners 4TWV0 and 4TTV0, acessado em junho 13, 2026, https://trane-mexico.com/productos/downloads/4TTV00.pdf
72. SL22KLV - Lennox, acessado em junho 13, 2026, https://www.lennox.com/dA/541bc6e8e0/100131.pdf
73. Superheat and sub cooling for unit with reversing valve : r/hvacadvice - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/hvacadvice/comments/1obaa09/superheat_and_sub_cooling_for_unit_with_reversing/
74. EEV not opening enough after defrost in walk-in refrigerator - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/refrigeration/comments/15bqw1e/eev_not_opening_enough_after_defrost_in_walkin/
75. iComfort S30 Installation Manual and Setup Guide - Lennox, acessado em junho 13, 2026, https://www.lennox.com/literature/Lennox_iComfortS30_Installation_Manual_Setup_Guide.pdf

76. Service and Troubleshooting - AC Direct, acessado em junho 13, 2026, https://www.acdirect.com/media/specs/Amana/amana-avpec-service-manual.pdf
77. What do you even check on these : r/HVAC - Reddit, acessado em junho 13, 2026, https://www.reddit.com/r/HVAC/comments/1dy90wf/what_do_you_even_check_on_these/
78. Digital vs Analog Manifold Gauges: A Comprehensive Comparison for HVAC Professionals, acessado em junho 13, 2026, https://hvacprosales.com/blog/digital-vs-analog-manifold-gauges-comparison/
79. Digital vs Analog Manifold Gauges: Efficiency Comparison in HVAC - ElitechEU, acessado em junho 13, 2026, https://www.elitecheu.com/blogs/elitecheu/digital-vs-analog-manifold-gauges-efficiency-comparison-in-hvac
80. Analog Gauge Set vs. Digital Manifold vs. Wireless Probes: Choosing the Right HVACR Tools - Fieldpiece, acessado em junho 13, 2026, https://www.fieldpiece.com/news-articles/analog-gauge-set-vs-digital-manifold-vs-wireless-probes-choosing-the-right-hvacr-tools/
81. HVAC Refrigerant P-T Chart - Apps on Google Play, acessado em junho 13, 2026, https://play.google.com/store/apps/details?id=com.hvactools.refrigerant_pt_chart&hl=en_US
82. Choosing the right HVAC-R manifold: analog or digital? - Iinstrument, acessado em junho 13, 2026, https://www.linstrument.com/en/blog/choosing-the-right-hvac-r-manifold-analog-or-digital
83. Heating Mode Performance Measurements For A Residential Heat Pump With Single-Faults Imposed - NIST Technical Series Publications, acessado em junho 13, 2026, https://nvlpubs.nist.gov/nistpubs/TechnicalNotes/NIST.TN.1648_2009.pdf
84. Cooling mode fault detection and diagnosis method for a residential heat pump - GovInfo, acessado em junho 13, 2026, https://www.govinfo.gov/content/pkg/GOVPUB-C13-d7b4b772d06bb6c15422f34c4b726194/pdf/GOVPUB-C13-d7b4b772d06bb6c15422f34c4b726194.pdf
85. 3161690 ADM TDS PT Chart Solstice 454B LTR, acessado em junho 13, 2026, https://prod-edam.honeywell.com/content/dam/honeywell-edam/pmt/oneam/en-us/refrigerants/documents/hon-ess-am-3161690-adm-tds-pt-chart-solstice-454b-ltr-en.pdf
86. R-410A Operating Pressures: Normal Ranges, Symptoms of Issues & How to Read, acessado em junho 13, 2026, https://www.acdirect.com/blog/r410a-operating-pressures/